Experimentation Challenges — A Field Guide

A well-designed A/B test can still produce the wrong answer. The challenges below represent the most common ways that measured lift diverges from true lift — through statistical errors, assumption violations, or behavioral artifacts. Each has a distinct root cause and a concrete solution.

01

Multiple Comparisons

Testing many metrics or variants simultaneously inflates the chance of finding a false positive — even when nothing truly changed.

↑ False positive rate

02

Peeking / Early Stopping

Reading results before the experiment reaches its planned sample size inflates false positive rates and makes results untrustworthy.

↑ False positive rate

03

Network Effects / Spillover

In connected networks, treated users influence control users — violating SUTVA and understating the true treatment effect.

↓ Understates true lift

04

Two-Sided Market Interference

In marketplaces, treated and control users compete for shared supply — deflating the control baseline and overstating lift.

↑ Overstates true lift

05

Novelty Effect

Existing users over-engage with new features simply because they are new — inflating short-term lift that decays over time.

↑ Overstates true lift

06

Simpson's Paradox

A trend visible in every segment can reverse in the aggregate when arms have unequal composition across a key dimension.

↕ Misleads aggregate result

Multiple Comparison - Problem Scenario

Problem Statement: Running more tests simultaneously increases chances of finding something significant by accident.

With α = 0.05, false positive risk grows with each additional test.

- with 1 test, false positive risk is 5%
  vs.
- with 5 tests, false positive risk is 23%

** false positive: incorrectly declaring a result significant when nothing real changed*

Tests	False Positive Risk
1	5%
2	10%
3	14%
4	19%
5	23%

Example: all primary and secondary metrics appear significant and the guardrail is safe — looks like a strong win, but our false positive risk is **23%**

<table>
<tr><th>Type</th><th>Metric</th><th>p-value</th><th>Significant? (no correction)</th><th>False Positive Risk</th></tr>
<tr><td>Primary</td><td>Revenue</td><td>0.012</td><td>✓ Yes</td><td rowspan="5">23%</td></tr>
<tr><td rowspan="3">Secondary</td><td>CTR</td><td>0.009</td><td>✓ Yes</td></tr>
<tr><td>Time Spent</td><td>0.028</td><td>✓ Yes</td></tr>
<tr><td>Page Views</td><td>0.045</td><td>✓ Yes</td></tr>
<tr><td>Guardrail</td><td>Opt-out Rate</td><td>0.080</td><td>✗ No (Safe)</td></tr>
</table>

Multiple Comparison - Correction Solutions

Solutions: To address multiple comparison, we adjust the significance threshold (α) for each test. There are 3 common approaches, choosing based on your business objective.

Method	Logic	Adjusted α (5 tests)	Best For
Bonferroni	Divide α equally	0.010 each	High stakes — multi-million dollar revenue calls, healthcare decisions, fraud detection
Holm-Bonferroni (HB)	Sequential adjustment	0.010, 0.013, 0.017, 0.025, 0.050	Most business experiments — product launches, campaign evaluation
Benjamini-Hochberg (BH)	Controls false discovery rate	0.010, 0.020, 0.030, 0.040, 0.050	Exploratory analysis — screening many features, early-stage hypothesis generation

Bonferroni Ex: High stakes decision → Primary not significant. Do not proceed.

HB Ex: Social supporting campaign → Primary significant, guardrail safe. Safe to continue.

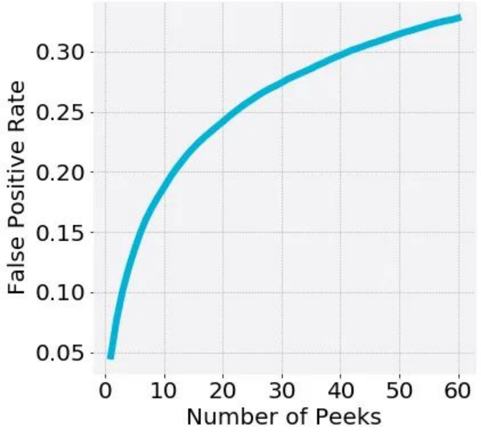
Type	Metric	p-value	No Correction	Bonferroni	HB	BH
Primary	Revenue	0.012	✓	✗	✓	✓
Secondary	CTR	0.009	✓	✓	✓	✓
	Time Spent	0.028	✓	✗	✗	✓
	Page Views	0.045	✓	✗	✗	✗
Guardrail	Opt-out Rate	0.080	✗ (Safe)	✗ (Safe)	✗ (Safe)	✗ (Safe)

** Correction methods applied in p-value rank order (smallest to largest)*

The Peeking Problem

What is peeking? Checking experiment results before the planned end date and acting on what you see — declaring a winner and stopping early.

Why is it a problem? This **inflates your false positive rate** — e.g., peek 5 times and your false positive risk is close to 10%, not 5%. The more you peek, the more likely you are to see a 'significant' result by chance.



Source: Skyscanner Engineering

Solutions

1. **Don't do it: Stick to the plan.**
   Let the experiment run to its pre-specified end date.

2. **Peek safely, if you must**: Purpose-built methods allow interim monitoring while keeping false positive rates in check.
   - **Alpha Spending**: Pre-commit to a fixed look schedule.
   - **Sequential Testing** (mSPRT): Peek any time.

Caveat: Safe peeking methods exist, but they set a much higher bar for early decisions.

The Peeking Problem - Alpha Spending Solution

When to Use: You know in advance when you'll check results.

How to Set it Up

1 Decide your look schedule

Choose # peek times and at what sample fractions, e.g. peek once at 50% of planned sample, once at 100%

2 Apply O'Brien-Fleming

Spends little alpha early and saves most of the budget for the final look. Overall error rate stays at 5%.

3 Compare at each look

Cross the threshold → stop early and declare a winner.
Don't cross it → continue to the next look

Learnings

- Early looks require a much stricter threshold. It takes an exceptionally strong signal to stop early.
- More looks make the final threshold harder to cross. **Two looks** is the industry sweet spot.

Ex: Look schedule	Look 1	Look 2	Look 3	Final look	Power cost
2 looks (50%, 100%)	$p < 0.005$	—	—	$p < 0.042$	Minimal ✓
3 looks (33%, 67%, 100%)	$p < 0.0006$	$p < 0.015$	—	$p < 0.034$	Moderate
4 looks (25%, 50%, 75%, 100%)	$p < 0.0001$	$p < 0.005$	$p < 0.017$	$p < 0.027$	Higher

Example: baseline 10%, MDE 2pp, α = 0.05, power = 80%, total sample 7,600

The Peeking Problem - mSPRT Solution

When to Use: A more flexible approach to peek any time, or MDE is uncertain at design time.

How it Works

- At every peek, compute a mixture likelihood ratio (mLR). The higher it is, the more the data points to a real effect.
- Stop as soon as mLR > 1/α (e.g., 20 for α = 0.05)
- The false positive guarantee holds at every peek.

Example: By peek 4, at just 80% of the planned sample, the mLR crosses 20, giving a statistically valid early stop.

Peek	Sample (n)	Control %	Test %	Z-stat	mLR	Decision
1	200 (20%)	20.1%	21.9%	0.69	2.21	Continue
2	400 (40%)	19.8%	22.2%	1.01	3.89	Continue
3	600 (60%)	20.2%	22.8%	1.38	7.14	Continue
4	800 (80%)	20.0%	23.5%	1.98	22.67 ✓	STOP — winner

Example: baseline 20%, α = 0.05, power = 80%, σ² = 0.01, total sample 1,000

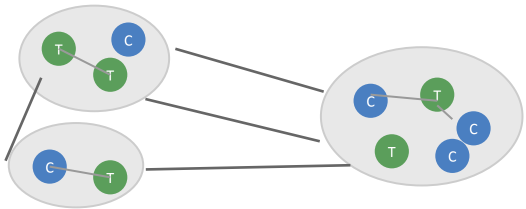
vs. SPRT: If your MDE is known upfront, SPRT stops faster when the true effect matches exactly. In practice, mSPRT is the better default — most business questions are "is there any effect?" rather than "did we hit a specific lift?"

Network Effect - Problem

Problem Statement: Standard A/B testing assumes users act independently (SUTVA). In connected networks, this breaks down: treated users influence control users through shared connections, causing treatment effects to leak across group boundaries and **understating the true lift**.

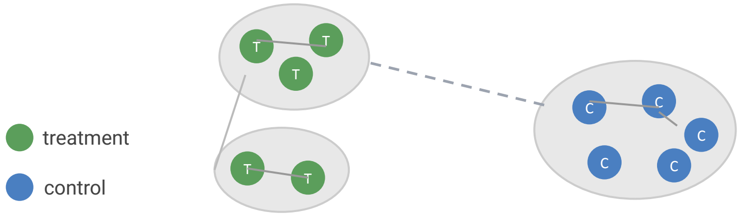
Unit Randomization

Mixed assignment — interference crosses group boundaries



SOLUTION: Cluster Randomization

Whole clusters assigned to one arm — cross-cluster edges minimized



Problem: Spillover

Control baseline inflates, measured lift understates true effect

Ex: Facebook / Instagram / TikTok / Linkedin

New feed algorithm → Treated user posts more →

Control friends see & engage → Control baseline inflates

Problem: Under-Treatment

Feature rarely fires, effect diluted at the source

Ex: WhatsApp

New product feature →

Feature requires both users in treatment to activate →

75% of treated users message control peers →

Dilutes the treatment experience

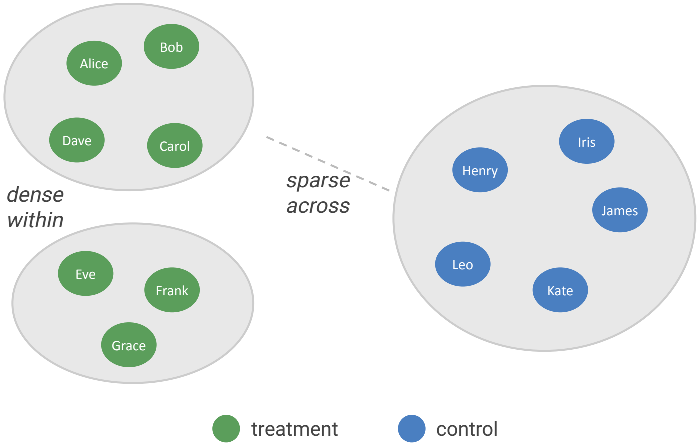
Network Effect - Solution

Cluster Randomization

- Use graph community detection (**Louvain**) to partition users into dense clusters with minimal cross-cluster edges.
- Randomize at the cluster level — spillover stays within arms, and treated users interact with other treated users

Method: Modularity Optimization (Louvain)

Algorithm: graph community detection



For each possible grouping, Louvain scores:

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Q  =  (actual edges within group)
     -(edges expected by chance)
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Higher Q = denser within-group connections. Louvain greedily merges nodes to maximize Q.

THE DESIGN TRADEOFF: POWER VS. PURITY

Goal: find the right cluster size for your product interaction density — enough purity to contain spillover, enough clusters to detect an effect.

More clusters, smaller size	Fewer clusters, larger size
+ More statistical power - More cross-cluster edges, more spillover risk	+ Cleaner containment, less spillover - Fewer independent units, wider confidence intervals

Two-Sided Market

Problem Statement: In two-sided marketplaces, buyers and sellers compete for **shared supply** — treating one user changes the resource pool available to everyone else, biasing the measured lift.

Ex: when Uber tests a new pricing feature, treated riders book faster and claim available drivers, leaving control riders with fewer options. Control experience degrades, with deflated baseline, and overstated lift.

Solutions: Three common solutions, each suited to different supply structures and experiment constraints.

1 Time-region switchback

Geo-local supply

Alternate treatment by **time window** and **region** — all users in a region share the same condition, eliminating cross-arm supply competition.

Uber, Lyft, DoorDash, Airbnb

→ switchback slide

2 Time-only switchback

National or platform-wide supply

Alternate treatment by **time window** — the entire platform flips between conditions across time windows.

E-commerce, national inventory platforms

→ switchback slide

3 Geo-based testing

Market-level launches

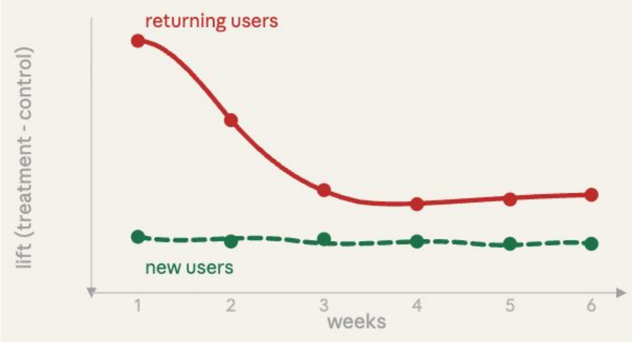
When user-level experimentation is infeasible — geo units isolate supply pools.

Matched markets: Pair similar Geos (DMAs), only when comparable markets exist → slide

Synthetic control: Weighted composite of untreated markets as counterfactual → slide

Novelty Effect

Problem Statement: When a new feature launches, existing users engage more than usual simply because it's new. This inflates experiment metrics during the test window. Once novelty wears off, engagement decays — and measured lift overstates the long-term true effect.



Run longer — read from the stable window

- Plot lift week over week. If still decaying, the experiment is not ready to read.
- Make launch decisions only from the stable window — not the overall average, which is inflated by the early novelty spike.

Segment by user tenure — use new users as the clean signal

- New users have no prior baseline, so their lift is uncontaminated by novelty.
- If returning users show high early lift that decays while new users stay stable, novelty is confirmed, then **use the new user** segment lift as long-term product value.

Simpson's Paradox

Problem Statement: A trend that appears consistently across individual segments can reverse — or disappear entirely — when those segments are combined. For example, one arm has disproportionately more users from a low-converting segment, dragging down the overall average.

Example: Control had far more Desktop users, inflating its aggregate. The composition difference drove the overall result

	Treatment	Control	Winner
Mobile Treatment: 800 users · Control: 200 users	**4.0%**	3.0%	Treatment ✓
Desktop Treatment: 200 users · Control: 800 users	**18.0%**	15.0%	Treatment ✓
Overall Treatment: 1,000 users · Control: 1,000 users	5.6%	**12.6%**	Control ✓

Solution

Prevent: Use stratified randomization to ensure key dimensions — device, market, user tenure — are balanced between arms by design, not by chance.

If detected:

- Trust segment-level results — they reflect the true treatment effect.
- If a single estimate is needed, reweight segment effects by true population distribution, not experimental composition.